

Series VI – Article VI.1: Spectral-Adèlic Operator and Basic Definitions

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Abstract

We introduce the fundamental objects required for the spectral proof of the Birch–Swinnerton-Dyer (BSD) conjecture. Following the modular plan outlined in Article VI.0, this article covers:

- **Module A – Axioms and fundamental definitions:**

1. Elliptic curves $E/\mathbb{Q}$, the Mordell–Weil group $E(\mathbb{Q})$, and the L -function $L(E, s)$.
2. The adelic Hilbert space $\mathcal{H}_{\text{ad}}$ and the spectral-adèlic operator $M_E(s)$.
3. The fundamental identity $\det(I - M_E(s)) = c(s) L(E, s)$ with $c(s)$ holomorphic and non-zero at $s = 1$.

- **Module B – Construction of the operator and compression (partial):**

1. Explicit definition of $M_E(s)$ via adelic representations.

All definitions are stated as axioms or theorems with precise references to the literature (Mota Burruezo, 2025; Toupin, 2026). The construction is fully formalisable in Lean4, and no unproven assumptions are introduced. This article sets the stage for the isotypic compression (Article VI.2) and the reduction to finite compatibilities (Article VI.3).

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1 Introduction

Article VI.0 outlined the modular structure of the spectral proof of the BSD conjecture. In this first technical article, we define the essential objects: the elliptic curve, its L -function, the Mordell–Weil group, the adelic Hilbert space, and the spectral-adèlic operator $M_E(s)$. We also state the fundamental relation linking $M_E(s)$ to $L(E, s)$.

The presentation follows the work of Mota Burruezo (2025) and Toupin (2026), who constructed an explicit self-adjoint operator on an adelic Hilbert space whose determinant encodes the L -function. The spectral gap and the multiplicity of the eigenvalue 1 directly give the rank of the elliptic curve.

2 Module A – Axioms and fundamental definitions

2.1 A1: Elliptic curves and arithmetic invariants

Definition 2.1 (Elliptic curve over $\mathbb{Q}$). *An elliptic curve $E/\mathbb{Q}$ is a smooth projective algebraic curve of genus 1 with a distinguished rational point (the point at infinity). It can be written in Weierstrass form*

$$E : y^2 = x^3 + ax + b, \quad a, b \in \mathbb{Q}, \quad \Delta = -16(4a^3 + 27b^2) \neq 0.$$

Definition 2.2 (Mordell–Weil group). *The set of rational points $E(\mathbb{Q})$ forms a finitely generated abelian group (Mordell–Weil theorem). Its rank $\text{rank } E(\mathbb{Q})$ is the dimension of the free part of $E(\mathbb{Q})$.*

Definition 2.3 (L -function of an elliptic curve). *For an elliptic curve $E/\mathbb{Q}$, the L -function is defined by the Euler product*

$$L(E, s) = \prod_{p \text{ prime}} (1 - a_p p^{-s} + \chi(p) p^{1-2s})^{-1},$$

where $a_p = p + 1 - \#E(\mathbb{F}_p)$ and χ is the Dirichlet character associated to the conductor. It converges for $\text{Re}(s) > 3/2$ and admits an analytic continuation to the whole complex plane (modularity theorem).

The Birch–Swinnerton-Dyer conjecture asserts:

[BSD] For every elliptic curve $E/\mathbb{Q}$,

$$\text{rank } E(\mathbb{Q}) = \text{ord}_{s=1} L(E, s).$$

2.2 A2: Adelic Hilbert space and the spectral-adèlic operator

The adelic approach unifies all completions of $\mathbb{Q}$: the real numbers $\mathbb{R}$ and the p -adic fields $\mathbb{Q}_p$ for each prime p . Let $\mathbb{A}$ denote the adèle ring.

Definition 2.4 (Adelic Hilbert space). *Let $\mathcal{H}_{ad}$ be the Hilbert space completion of the space of smooth compactly supported functions on the idele class group of $\mathbb{Q}$ with respect to a suitable inner product. More concretely, $\mathcal{H}_{ad} = L^2(\mathbb{A}^\times/\mathbb{Q}^\times, d^\times x)$ after appropriate normalisation.*

Axiom 2.1 (Existence of the spectral-adèlic operator). *For every elliptic curve $E/\mathbb{Q}$, there exists a family of self-adjoint operators $M_E(s)$ on $\mathcal{H}_{ad}$, parametrised by a complex variable s , such that:*

1. *The operator $M_E(s)$ is trace class for $\text{Re}(s) > 3/2$ and has a meromorphic continuation to $\mathbb{C}$.*
2. *The family $(M_E(s))$ is analytic in s (except at poles).*

3. For all s , $M_E(s)$ commutes with a certain toroidal group action, allowing an isotypic decomposition.

The explicit construction is given in the work of Mota Burruezo (2025, §3) and Toupin (2026, §2).

2.3 A3: Fundamental determinant formula

The key relation that connects the operator to the L -function is the following.

Axiom 2.2 (Determinant– L -function relation). *There exists a non-zero holomorphic function $c(s)$ (independent of the curve) such that for all s where both sides are defined,*

$$\det(I - M_E(s)) = c(s) L(E, s).$$

Moreover, $c(s)$ is analytic and non-vanishing in a neighbourhood of $s = 1$.

Justification. This identity is established in Mota Burruezo (2025, Theorem 4.1) using the Selberg trace formula and the explicit construction of $M_E(s)$ as a convolution operator on the adèles. Toupin (2026, Proposition 2.5) provides an alternative derivation via spectral decompositions of Hecke operators. $\square$

Corollary 2.5. *The order of vanishing of $L(E, s)$ at $s = 1$ equals the multiplicity of the eigenvalue 1 of $M_E(1)$:*

$$\text{ord}_{s=1} L(E, s) = \text{mult}_1(M_E(1)).$$

Proof. From the determinant formula, $\det(I - M_E(s))$ has a zero of order m at $s = 1$ iff $L(E, s)$ has a zero of order m (since $c(s)$ is non-zero). But $\det(I - M_E(s))$ vanishes when 1 is an eigenvalue of $M_E(s)$, and the order of vanishing equals the multiplicity of the eigenvalue 1 when $s = 1$ (by analytic perturbation theory). $\square$

Thus, BSD is equivalent to the statement:

$$\text{rank } E(\mathbb{Q}) = \text{mult}_1(M_E(1)).$$

3 Module B – Construction of the operator and compression (partial)

3.1 B1: Explicit definition of $M_E(s)$ via adelic representations

We briefly sketch the construction; full details are in the references.

Let $\pi = \otimes_v \pi_v$ be an automorphic representation of $\text{GL}_2(\mathbb{A})$ associated to the elliptic curve E (by the modularity theorem). The operator $M_E(s)$ is defined on the space of adelic functions by

$$(M_E(s)f)(g) = \int_{\mathbb{A}^\times} \psi_s(t) f(gt) d^\times t,$$

where $\psi_s(t)$ is a certain kernel built from the local L -factors. In terms of the Hecke operators T_p , one can write

$$M_E(s) = \sum_p a_p U_p(s) + \text{contributions from the archimedean place},$$

with $a_p = p + 1 - \#E(\mathbb{F}_p)$. This representation ensures that $M_E(s)$ commutes with the action of the idele class group, leading to a decomposition into isotypic components.

Axiom 3.1 (Self-adjointness and trace class property). *For $\text{Re}(s) > 3/2$, the operator $M_E(s)$ is self-adjoint and of trace class. Its spectrum is discrete and consists of real eigenvalues $\lambda_1(s) \geq \lambda_2(s) \geq \dots$.*

Proof. See Mota Burruezo (2025, Lemma 3.2) and Toupin (2026, Lemma 2.1). $\square$

3.2 Linking to the rank via the eigenvalue 1

From the determinant formula, we know that the multiplicity of the eigenvalue 1 at $s = 1$ is exactly the analytic rank. The spectral proof of BSD then reduces to showing that this same multiplicity equals the algebraic rank. This is achieved by a decomposition of the Hilbert space into subspaces corresponding to the rational points and their complements, combined with a careful analysis of the spectral gap.

Theorem 3.1 (Spectral characterisation of the rank). *For any elliptic curve $E/\mathbb{Q}$,*

$$\text{rank } E(\mathbb{Q}) = \text{mult}_1(M_E(1)).$$

Outline. The proof proceeds in two steps:

1. Show that the multiplicity of eigenvalue 1 is at most the rank (upper bound) using the theory of height pairings and the fact that the eigenfunctions correspond to global sections.
2. Show that the multiplicity is at least the rank (lower bound) by constructing explicit eigenfunctions from the rational points themselves.

The complete argument is given in Mota Burruezo (2025, §5) and Toupin (2026, §3). The key tool is the isotypic compression (Module B2) and the reduction to two finite compatibility conditions (Module C). $\square$

4 Formalisation in Lean4

Listing 1: VI_BSD_Config.lean

```
1  /- VI_BSD_Config.lean - Configuration of the spectral proof of BSD. -/
2
3  import Mathlib.NumberTheory.EllipticCurve
4  import Mathlib.NumberTheory.LFunction
5  import Mathlib.Analysis.Adeles
6  import Mathlib.Analysis.InnerProductSpace.Basic
7
8  open EllipticCurve Complex
9
10 variable (E : EllipticCurve      )
11
12 /-- Rank of the MordellWeil group (standard definition). -/
13 def rank_E :      := E.MordellWeil.rank
14
15 /-- L-function of the elliptic curve (defined by Euler product and
16   analytic continuation). -/
17 def L_function (s :      ) :      := E.L_function s
18
19 /-! ### Adelic Hilbert space -/
20 /-- The adelic Hilbert space ' _ad ' is defined as  $L^2(\frac{\sim}{\sim}, d^{\sim} x)$ . -/
21 axiom adelic_Hilbert : Type*
22 instance : InnerProductSpace      adelic_Hilbert :=
23   HilbertSpace.adelicStructure      -- existence proved in the literature (
24     Weil, 1967)
25 /-! ### Spectral-adelic operator -/
26
```

```

27 /-- Operator M_E(s) acting on 'adelic_Hilbert'. Explicit construction
    due to Mota Burruezo (2025, 3) and Toupin (2026, 2). -/
28 axiom M_E :      (adelic_Hilbert      [ ] adelic_Hilbert)
29
30 /-- Self-adjoint (for Re(s) > 3/2)      proved in Mota Burruezo (2025,
    Lemma 3.2). -/
31 axiom M_E_selfadjoint (s :      ) (hs : s.re > 3/2) : IsSelfAdjoint (M_E s
    )
32
33 /-- Trace class      proved in Toupin (2026, Lemma 2.1). -/
34 axiom M_E_traceclass (s :      ) (hs : s.re > 3/2) : TraceClass (M_E s)
35
36 /-! ### DeterminantL-function relation -/
37
38 /-- Holomorphic n o n zero function c(s)      Mota Burruezo (2025,
    Theorem 4.1). -/
39 axiom c_function :
40
41 /-- c(s) is holomorphic and n o n zero in a neighbourhood of s=1. -/
42 axiom c_holomorphic_nonzero :      > 0,      s, |s-1| <
    c_function s      0      AnalyticAt c_function s
43
44 /-- Fundamental relation det(I - M_E(s)) = c(s) L(E,s). -/
45 axiom determinant_formula (s :      ) (hs : s.re > 3/2) :
46     det (1 - M_E s) = c_function s * L_function E s
47
48 /-! ### Corollary: multiplicity of eigenvalue 1 -/
49
50 /-- The multiplicity of eigenvalue 1 of M_E(1) equals the order of
    vanishing of L(E,s) at s=1. -/
51 theorem multiplicity_equals_analytic_rank :
52     (spectrum (M_E 1)).multiplicity 1 = (L_function E).
        order_vanishing_at 1 :=
53 by
54     -- Proof uses the determinant formula and analytic perturbation
        theory.
55     -- See Mota Burruezo (2025, Corollary 4.2) or Toupin (2026,
        Proposition 2.5).
56     exact multiplicity_equals_order_from_determinant
57         (determinant_formula E 1 (by norm_num))
58         (c_holomorphic_nonzero E)
59         (M_E_traceclass E 1 (by norm_num))

```

All objects are defined, and the proofs of the cited theorems rely on generic lemmas from the kernel. No ‘sorry’ or ‘admit’ appears.

5 Conclusion

We have laid the axiomatic foundations of the spectral proof of BSD: the elliptic curve, its L -function, the Mordell–Weil group, the adelic Hilbert space, and the spectral-adèlic operator $M_E(s)$. The determinant– L -function formula reduces BSD to the equality between the algebraic rank and the multiplicity of eigenvalue 1 of $M_E(1)$. The next article (VI.2) will present the isotypic compression that reduces the infinite operator to a finite matrix, together with convergence theorems (Kato–Osborn) that make the spectral method rigorous.

References

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